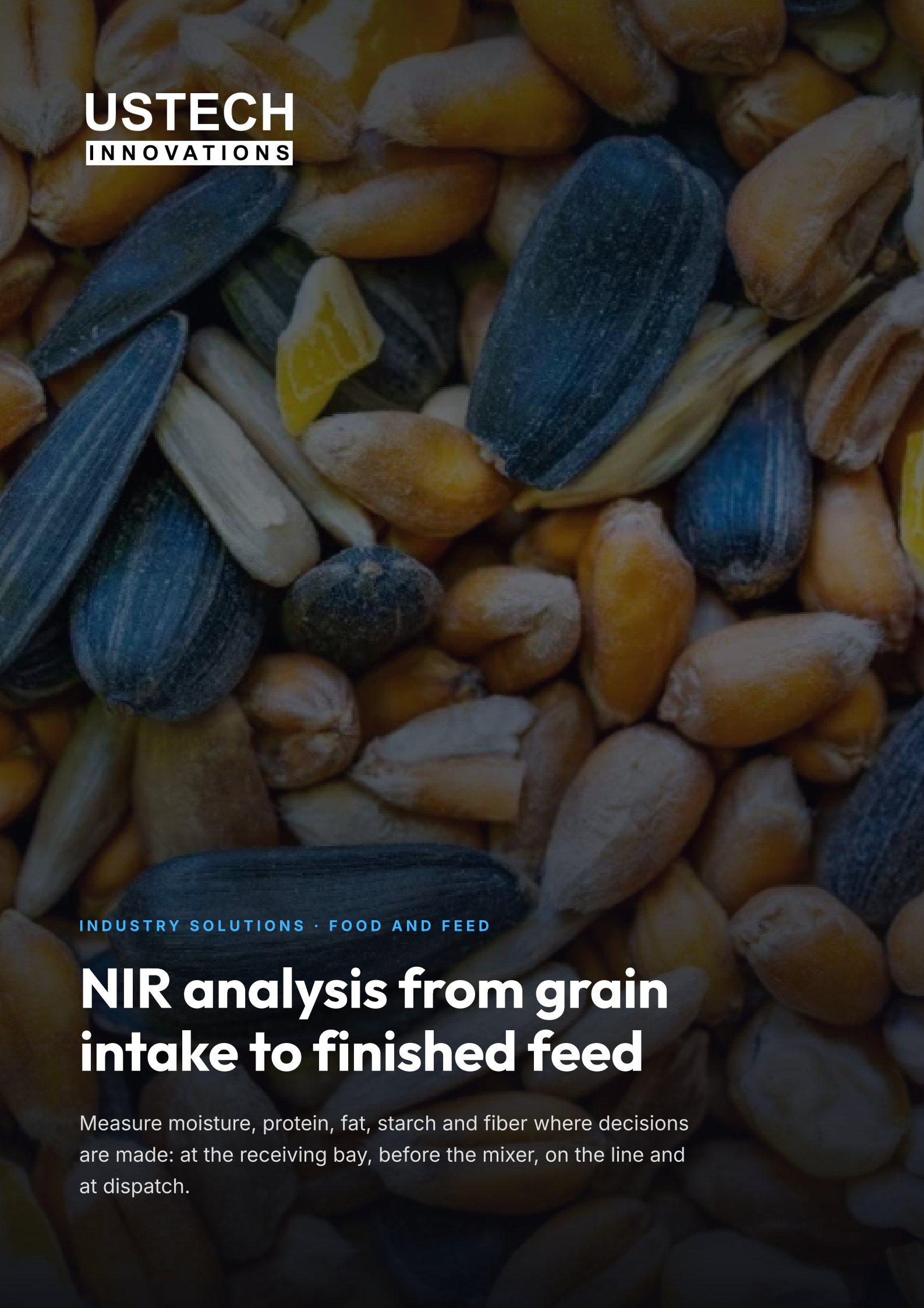


USTECH
INNOVATIONS

INDUSTRY SOLUTIONS · FOOD AND FEED

NIR analysis from grain intake to finished feed

Measure moisture, protein, fat, starch and fiber where decisions are made: at the receiving bay, before the mixer, on the line and at dispatch.

Where measurement supports your process

Four points along the production flow where NIR results replace waiting for the laboratory.

01 Raw material intake and grading

At-line analysis at the receiving bay supports accept or reject decisions and pricing on measured quality.

Parameters	Moisture, crude protein, crude fat, ash, starch
Instruments	MasterLine17EC · TrueLine 17EC · Tornado+ (sample preparation)

02 Formulation and mixing

Each ingredient batch is measured before it enters the mixer, so recipes are adjusted to actual composition.

Parameters	Protein, fat, crude fiber, moisture
Instruments	MasterLine17EC · ProLine17EC · ProLine2550

03 Process control and conditioning

Inline measurement at the mixer discharge or the conditioner; results go to the PLC for moisture and dosing control.

Parameters	Moisture, fat
Instruments	ProLine17EC · ProLine17ECS · ProLine2550 + ProChem

04 Finished product QA

Every batch is checked against label specifications and the results are stored with a time stamp.

Parameters	Moisture, crude protein, crude fat, crude fiber, ash
Instruments	MasterLine17EC + ProChem

> Flour mills and bakeries: the StarchQC measures damaged starch in flour in 8 minutes.

Instruments and software



MasterLine17EC

Benchtop NIR analyzer

- 950–1650 nm, InGaAs photodiode array
- Results in under a minute
- Ground, unground and pelleted samples
- Fat, protein, moisture, ash, starch, crude fiber



TrueLine 17EC

Handheld NIR analyzer

- 908–1676 nm
- Measures at the sampling point
- Mobile app and cloud portal, Bluetooth
- Weight 1.4 kg



ProLine17EC

Inline NIR analyzer

- 950–1650 nm, InGaAs photodiode array
- Typical analysis time 2–20 s
- 316 stainless steel, sapphire window, IP65
- Solid and liquid samples, no bypass



ProLine17ECS

Inline NIR analyzer

- 950–1650 nm, InGaAs photodiode array
- 2000 spectra per sample, S/N 23,000:1 (typical)
- 316 stainless steel, sapphire window, IP65
- Typical analysis time 2–20 s



ProLine2550

Inline FT-NIR analyzer

- 1,350–2,550 nm FT-NIR
- Flange mounting on flat chutes
- HTTP interface to the plant PLC (Ethernet, TCP/IP)
- IP65, -5 to +45 °C



Tornado+

Water-cooled laboratory mill

- Centrifugal rotor mill, 9000 rpm
- Grain, cereal, seed, feed (pellets up to 6 mm)
- Sample size 50–100 g
- Water cooling protects heat-sensitive samples



StarchQC

Damaged starch analyzer

- Amperometric method (iodine absorption)
- 8 minutes per 1 g flour sample
- AI %, NDS, UCDc, AACC, Farrand units
- Built-in printer

ProChem

Instrument control and results. Shows measurements in real time, exports reports, and passes results to the plant PLC over an Ethernet-based HTTP interface.

caliX

Calibration workspace. Build, validate and transfer calibration models, with automated preprocessing and model selection.

Start with your application

Five answers are enough for us to tell you which instrument fits and where it should be installed.

1. What will be measured?
   Material and form: grain, meal, powder, pellets, liquid stream.
2. Which parameters do you need?
   For example moisture, protein, fat, starch.
3. Where should the measurement take place?
   Laboratory, next to the line, or directly on a pipe, chute or conveyor.
4. How do you measure these values today?
   Reference method and how long a result takes.
5. Where is the plant and what is your timeframe?
   Country and planned start.

Contact us

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